

Task 1

To Change Directories, Validate Paths, Update Working Directory, Handle Errors, Maintain Previous Directory, Test Navigation

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <limits.h>
#include <sys/stat.h>

int main() {
    char path[PATH_MAX];
    char current_dir[PATH_MAX];
    char previous_dir[PATH_MAX] = "";
    struct stat st;

    printf("Simple Directory Navigation Program\n");
    printf("Commands: cd <path>, pwd, previous, exit\n\n");

    while (1) {
        if (getcwd(current_dir, sizeof(current_dir)) == NULL) {
            perror("getcwd");
            break;
        }
        printf("%s$ ", current_dir);

        if (fgets(path, sizeof(path), stdin) == NULL) {
            break;
        }
        path[strcspn(path, "\n")] = '\0';

        if (strcmp(path, "exit") == 0) {
            printf("Exiting...\n");
            break;
        }
        if (strcmp(path, "pwd") == 0) {
            printf("Current Directory: %s\n", current_dir);
            continue;
        }
        if (strcmp(path, "previous") == 0) {
            if (strlen(previous_dir) == 0) {
                printf("No previous directory available.\n");
                continue;
            }
            char temp[PATH_MAX];
            strcpy(temp, current_dir);
            if (chdir(previous_dir) == -1) {
                perror("chdir");
                continue;
            }
            strcpy(previous_dir, temp);
            printf("Changed to previous directory.\n");
            continue;
        }
        if (strncmp(path, "cd ", 3) == 0) {
            char *new_path = path + 3;
            while (*new_path == ' ') {
                new_path++;
            }
            if (stat(new_path, &st) == -1) {
                perror("Invalid path");
                continue;
            }
            if (!S_ISDIR(st.st_mode)) {
                printf("Error: Path is not a directory.\n");
                continue;
            }
            strcpy(previous_dir, current_dir);
            if (chdir(new_path) == -1) {
                perror("chdir");
                continue;
            }
            if (getcwd(current_dir, sizeof(current_dir)) == NULL) {
                perror("getcwd");
                continue;
            }
            printf("Directory changed successfully.\n");
            printf("New Working Directory: %s\n", current_dir);
        } else {
            printf("Invalid command.\n");
            printf("Use: cd <path>, pwd, previous, or exit\n");
        }
    }
    return 0;
}
```

OUTPUT:

```
imvarls@LAPTOP-VQ4D70L7: × + ∨  
imvarls@LAPTOP-VQ4D70L7:~$ nano navigation.c  
imvarls@LAPTOP-VQ4D70L7:~$ gcc navigation.c -o navigation  
imvarls@LAPTOP-VQ4D70L7:~$ ./navigation  
Simple Directory Navigation Program  
Commands: cd <path>, pwd, previous, exit  
  
/home/imvarls$ pwd  
Current Directory: /home/imvarls  
/home/imvarls$ |
```

Task 2

To Identify Built-ins, Create Dispatch Table, Execute In Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic

```
imvarls@LAPTOP-VQ4D70L7: × + v
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

#define MAX_INPUT 100
#define MAX_ARGS 20

void command_pwd(char **args);
void command_cd(char **args);
void command_help(char **args);
void command_exit(char **args);

typedef struct {
    char *name;
    void (*function) (char **args);
} Command;

Command dispatch_table[] = {
    {"pwd", command_pwd},
    {"cd", command_cd},
    {"help", command_help},
    {"exit", command_exit}
};

int command_count = sizeof(dispatch_table) / sizeof(dispatch_table[0]);

void command_pwd(char **args) {
    char cwd[1024];
    if (getcwd(cwd, sizeof(cwd)) != NULL) {
        printf("%s\n", cwd);
    }
}

void command_cd(char **args) {
    if (args[1] == NULL) {
        printf("cd: missing argument\n");
        return;
    }
    if (chdir(args[1]) != 0) {
        perror("cd");
    }
}

void command_help(char **args) {
    printf("\nAvailable commands:\n");
    printf(" pwd      Show current directory\n");
    printf(" cd <dir> Change directory\n");
    printf(" help      Show available commands\n");
    printf(" exit      Exit shell\n");
    printf(" Other commands are executed as processes.\n\n");
}
```

```

GNU nano 8.7.1 shell.c
int execute_builtin(char **args) {
    for (int i = 0; i < command_count; i++) {
        if (strcmp(args[0], dispatch_table[i].name) == 0) {
            dispatch_table[i].function(args);
            return 1;
        }
    }
    return 0;
}

int parse_input(char *input, char **args) {
    int count = 0;
    char *token = strtok(input, " \t\n");
    while (token != NULL && count < MAX_ARGS - 1) {
        args[count++] = token;
        token = strtok(NULL, " \t\n");
    }
    args[count] = NULL;
    return count;
}

int main() {
    char input[MAX_INPUT];
    char *args[MAX_ARGS];

    printf("=====\n");
    printf(" SIMPLE COMMAND SHELL\n");
    printf("=====\n");

    while (1) {
        fflush(stdout);
        printf("myshell> ");

        if (fgetc(input, sizeof(input), stdin) == NULL) {
            break;
        }
        if (strlen(input) <= 1) {
            continue;
        }

        int argc = parse_input(input, args);
        if (argc == 0) {
            continue;
        }
        if (execute_builtin(args)) {
            continue;
        }
        execute_process(args);
    }
    return 0;
}

```

OUTPUT:

```

imvaris@LAPTOP-VQ4D70L7: ~$ pmd
home/imvaris
imvaris@LAPTOP-VQ4D70L7:~$ ls
file.c  OSGP  copy.c  file.c  filename.c  myshell  process  processstate.c  shell  task2.c  week4.c  week5task2.c  week6server.c  zombie
file1.tx  client  copy_file.c  file.java  filename.c.save  myshell.c  process.c  server  shell.c  week1  week4  week5client  week6task2  zombie.c
file1.txt  client.c  demo  file2.c  first.c.save  navigation  process_state  server.c  snap  week1.c  week5.c  week6client.c  week6task2.c
file1.txt.c  copy  filename  first.c.save.1  navigation.c  process_state.c  server  task2  week4  week5task2  week6server  week6task2

imvaris@LAPTOP-VQ4D70L7:~$ help
GNU bash, version 5.3.9(1)-release (x86_64-pc-linux-gnu)
These shell commands are defined internally. Type 'help' to see this list.
Type 'help name' to find out more about the function 'name'.
Use 'info bash' to find out more about the shell in general.
Use 'man -k' or 'info' to find out more about commands not in this list.

star (*) next to a name means that the command is disabled.

! PIPELINE
job_spec [g]
(( expression ))
[ -p path ] filename [arguments]
[ arg... ]
[[ expression ]]
alias [-p] [name=value] ... ]
bg [job_spec ...]
bind [-lsvPSWX] [-m keymap] [-f filename] [-q name] [-u name] [-r keyseq] [-x keyseq:shell-command] >
break [n]
builtin [shell-builtin [arg ...]]
caller [exp]
case WORD in [PATTERN]...) COMMANDS ;;)... esac
cd [-l[[-P [-a]]] [-@] [dir]
command [-pv] command [arg ...]
compgen [-v varname] [-abcedfgjkuv] [-o option] [-A action] [-G globpat] [-W wordlist] [-F function]>
complete [-abcedfgjkuv] [-pr] [-DEI] [-o option] [-A action] [-G globpat] [-W wordlist] [-F function]>
compept [-o]o option] [-DEI] [name ...]
continue [n]
coproc [NAME] command [redirections]
declare [-aAffgillnrtux] [name=value] ...] or declare -p [-aAffgillnrtux] [name ...]
dirs [-clpu] [+H] [+d]
disown [-h] [-ar] [jobspec ... | pid ...]
echo [-neE] [arg ...]
enable [-a] [-dnps] [-f filename] [name ...]
eval [arg ...]
exec [-cl] [-a name] [command [argument ...]] [redirection ...]
exit [n]
export [-fn] [name=value] ...] or export -p [-f]
false
fc [-e ename] [-lnr] [first] [last] or fc -s [pat=rep] [command]
fg [job_spec]
for NAME [in WORDS ... ] ; do COMMANDS; done
for (( exp1; exp2; exp3 )); do COMMANDS; done
function name { COMMANDS ; } or name () { COMMANDS ; }
getopts optstring name [arg ...]
hash [-l] [-p pathname] [-dt] [name ...]
help [-dms] [pattern ...]

```

Task 3:

To Retrieve Current Directory, Display Path, Process Exit Requests, Cleanup Resources, Save State, Verify Correct Behavior

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <limits.h>
#include <string.h>

void cleanup() {
    printf("\nCleaning up resources...\n");
    printf("Resources cleaned successfully.\n");
}

int main() {
    char cwd[PATH_MAX];
    atexit(cleanup);

    printf("=== Directory and Process Management ===\n\n");

    if (getcwd(cwd, sizeof(cwd)) != NULL) {
        printf("Current Directory: %s\n", cwd);
    } else {
        perror("getcwd");
        return 1;
    }

    printf("Path: %s\n", cwd);

    FILE *file = fopen("state.txt", "w");
    if (file != NULL) {
        fprintf(file, "Saved Working Directory: %s\n", cwd);
        fclose(file);
        printf("State saved successfully.\n");
    } else {
        perror("State file");
    }

    if (getcwd(cwd, sizeof(cwd)) != NULL) {
        printf("Verification: Current directory is correct.\n");
    } else {
        printf("Verification failed.\n");
    }

    char choice;
    printf("\nDo you want to exit? (y/n): ");
    scanf(" %c", &choice);

    if (choice == 'y' || choice == 'Y') {
        printf("Process exit requested.\n");
        exit(0);
    } else {
        printf("Process will continue running.\n");
        return 0;
    }
}
```

OUTPUT:

```
imvarls@LAPTOP-VQ4D70L7:~$ nano directory.c
imvarls@LAPTOP-VQ4D70L7:~$ gcc directory.c -o directory
imvarls@LAPTOP-VQ4D70L7:~$ ./directory
=== Directory and Process Management ===

Current Directory: /home/imvarls
Path: /home/imvarls
State saved successfully.
Verification: Current directory is correct.

Do you want to exit? (y/n): y
Process exit requested.

Cleaning up resources...
Resources cleaned successfully.
imvarls@LAPTOP-VQ4D70L7:~$ |
```

Task 4

To Parse Export Syntax, Update Environment Variables, Validate Names, Handle Existing Variables, Export to Child Processes, Test Functionality

```
imvarls@LAPTOP-VQ4D70L7: × + ∨
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>
#include <unistd.h>
#include <sys/wait.h>

#define MAX_INPUT 200
#define MAX_NAME 100
#define MAX_VALUE 100

int valid_name(char *name) {
    if (name == NULL || name[0] == '\0')
        return 0;
    if (!isalpha(name[0]) && name[0] != '_')
        return 0;
    for (int i = 1; name[i] != '\0'; i++) {
        if (!isalnum(name[i]) && name[i] != '_')
            return 0;
    }
    return 1;
}

void export_variable(char *input) {
    char name[MAX_NAME];
    char value[MAX_VALUE];
    char *assignment = input + 7;
    char *equal = strchr(assignment, '=');

    if (equal == NULL) {
        printf("Error: Invalid export syntax.\n");
        printf("Usage: export NAME=VALUE\n");
        return;
    }

    int name_length = equal - assignment;
    if (name_length <= 0 || name_length >= MAX_NAME) {
        printf("Error: Invalid variable name.\n");
        return;
    }

    strncpy(name, assignment, name_length);
    name[name_length] = '\0';
    strcpy(value, equal + 1);

    if (!valid_name(name)) {
        printf("Error: Invalid variable name: %s\n", name);
        return;
    }

    char *old_value = getenv(name);
    if (old_value != NULL) {
        printf("Existing variable found: %s=%s\n", name, old_value);
    }

    if (setenv(name, value, 1) != 0) {
        perror("setenv");
        return;
    }
}
```

GNU nano 8.7.1

```
        return;
    }
    printf("Export successful: %s=%s\n", name, value);
}

void test_child_process() {
    pid_t pid = fork();
    if (pid < 0) {
        perror("fork");
        return;
    }
    if (pid == 0) {
        printf("\nChild Process: \n");
        char *value = getenv("MYVAR");
        if (value != NULL)
            printf("MYVAR=%s\n", value);
        else
            printf("MYVAR is not set.\n");
        exit(0);
    } else {
        wait(NULL);
        printf("Child process completed.\n");
    }
}

int main() {
    char input[MAX_INPUT];
    printf("=====\n");
    printf(" EXPORT COMMAND TESTER\n");
    printf("=====\n");
    while (1) {
        printf("\nshell> ");
        if (fgets(input, sizeof(input), stdin) == NULL)
            break;
        input[strcspn(input, "\n")] = '\0';
        if (strcmp(input, "exit") == 0) {
            printf("Exiting...\n");
            break;
        }
        if (strncmp(input, "export ", 7) == 0) {
            export_variable(input);
        } else if (strcmp(input, "test") == 0) {
            test_child_process();
        } else if (strncmp(input, "print ", 6) == 0) {
            char *name = input + 6;
            char *value = getenv(name);
            if (value != NULL)
                printf("%s=%s\n", name, value);
            else
                printf("%s is not set.\n", name);
        } else {
            printf("Unknown command.\n");
        }
    }
    return 0;
}
```


OUTPUT:

```
imvarls@LAPTOP-VQ4D70L7: × + v
imvarls@LAPTOP-VQ4D70L7:~$ nano export.c
imvarls@LAPTOP-VQ4D70L7:~$ gcc export.c -o export
imvarls@LAPTOP-VQ4D70L7:~$ ./export
=====
EXPORT COMMAND TESTER
=====

shell> MYVAR=Hello
Unknown command.

shell> test

Child Process:
MYVAR is not set.
Child process completed.

shell> |
```